Real-Time Social Media Analytics on the Israel-Palestine Conflict

1. Introduction

In the age of social media, public sentiment and engagement can be monitored in real-time, providing valuable insights into ongoing issues. This project focuses on building a dynamic, real-time dashboard that visualizes social media data surrounding the Israel-Palestine conflict. The goal is to help communications teams, researchers, and advocates monitor social media activity, identify trending topics, and analyze sentiment and engagement levels with respect to the conflict. The dashboard updates every second and provides various visualizations, such as sentiment analysis, emotional pulse, trending topics, geographic sentiment distribution, and topic engagement analysis.

2. Project Overview and Problem Statement

The Israel-Palestine conflict has been a highly debated topic on social media, with posts ranging from reports of violence to calls for peace. The sheer volume and diversity of opinions make it difficult for stakeholders to monitor real-time sentiment and engagement without the aid of automated tools. Existing methods of analysis often require manual efforts or are not updated in real-time, which limits the effectiveness of decision-making in communication efforts.

This project aims to address this gap by providing a real-time, interactive dashboard that displays key social media metrics, including sentiment polarity, engagement (likes, retweets, replies), and geographic distribution. By offering a dynamic, real-time view of public discourse, this dashboard will enable teams to respond to trends and public sentiment effectively.

3. Goals and Objectives

The main goals of the project were:

- To build a real-time dashboard that visualizes key social media metrics such as sentiment polarity, engagement rate, and trending topics.
- To allow filtering of data based on topic, emotion, and sentiment, enabling users to drill down into specific aspects of the discourse.
- To provide insight into the geographic distribution of sentiment and engagement, helping to understand regional sentiment.
- To make the dashboard interactive and continuously updated to reflect the most recent data.

4. Data Collection and Methodology

For the purpose of this project, real tweet data was not sourced directly from Twitter but was simulated using a custom Python script. The script generates tweet-like data based on a set of predefined topics and sentiment values that are common in discussions surrounding the Israel-Palestine conflict. The data generated includes the following attributes:

- **Tweet ID**: A unique identifier for each tweet.
- **Created At**: Timestamp of when the tweet was created.
- **Text**: Content of the tweet, including the topic and hashtag.
- **Sentiment Polarity**: The sentiment of the tweet, ranging from -1 (negative) to 1 (positive).
- **Emotion**: The emotion expressed in the tweet, including options like "anger," "hope," "neutral," etc.
- **Engagement Metrics**: Retweets, replies, and likes associated with the tweet.
- **Geolocation**: Latitude and longitude of the user who posted the tweet.
- **Is Trending**: Boolean value indicating whether the tweet is part of a trending topic.

The data was stored in a session state and updated every second. A queue was used to maintain the last 5 minutes of data, ensuring that the dashboard always displays the most recent activity.

5. Technology Stack

The dashboard was developed using the following technologies:

- **Streamlit**: For building the real-time interactive web application. Streamlit allows rapid prototyping of dashboards with minimal code.
- **Plotly**: For creating interactive, high-quality visualizations. Plotly is used to render charts and maps in the dashboard.
- **Python**: The primary programming language used for data generation, processing, and handling the dashboard logic.
- **Pandas**: For managing and processing the tweet data.
- **Geographical Libraries**: To visualize sentiment distribution on a map based on geographic coordinates (latitude and longitude).

6. Dashboard Design and Features

The dashboard has been designed to provide a comprehensive, real-time view of the social media engagement surrounding the Israel-Palestine conflict. The design focuses on clarity, interactivity, and real-time updates. Below are the key features:

- **Key Metrics Section**: Displays vital statistics like Sentiment Pulse, Engagement Rate, Trending Topics, and Peak Engagement. These metrics provide a quick snapshot of the ongoing social media activity.
- **Real-Time Analytics**: A line chart visualizing sentiment polarity and engagement over time, allowing users to observe fluctuations in public sentiment and interactions.
- **Trending Topics**: A table that lists the latest trending topics, along with their associated sentiment (positive, negative, or neutral).
- **Emotion Distribution**: A pie chart that breaks down the emotional tone of the tweets into categories like hope, anger, sadness, etc. This provides insight into the overall emotional response to the conflict.
- **Topic Engagement Analysis**: A bar chart showing which topics (e.g., conflict updates, peace talks, human rights) are generating the most engagement. This helps identify which aspects of the conflict are most discussed.
- **Geographic Sentiment and Engagement**: An interactive map showing the sentiment and engagement levels across different regions. This visualization helps to understand regional differences in the social media discourse.

7. Data Processing and Real-Time Updates

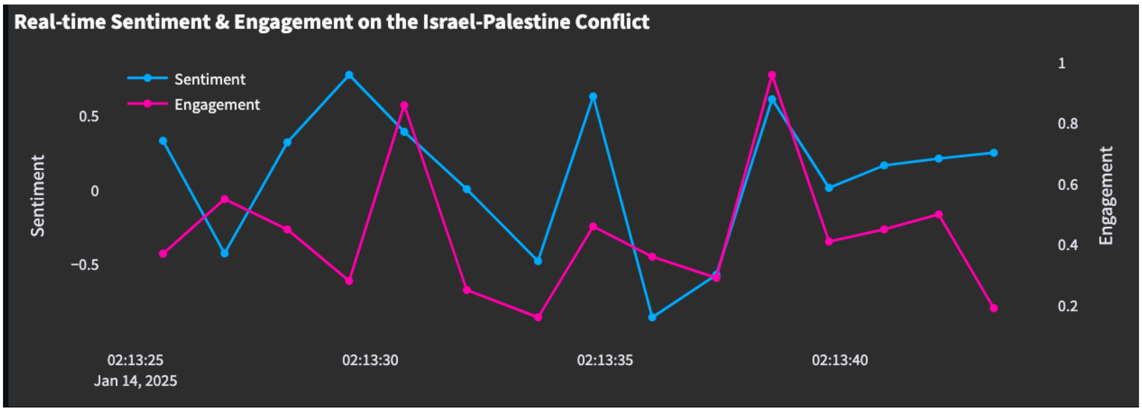
The real-time nature of the dashboard is achieved by generating new tweet data every second and updating the session state accordingly. The generate_realistic_tweet() function simulates a new tweet by:

- Selecting a topic based on time-of-day weighted probabilities.
- Generating sentiment using a Gaussian distribution.
- Adding engagement metrics like retweets, replies, and likes, based on the chosen topic and time.

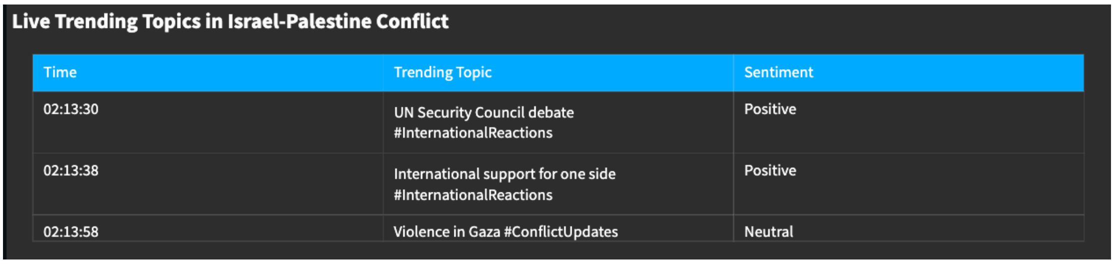
Once a new tweet is generated, it is added to a data queue. The dashboard continually updates by retrieving the latest data from the queue and recalculating key metrics like average sentiment and total engagement. This ensures that the dashboard always shows the most current state of social media activity.

8. Visualizations: Several types of visualizations were created to help users explore the data

- **Real-Time Sentiment & Engagement**: This line chart shows sentiment and engagement over time. The sentiment is plotted on one axis, while engagement (normalized to a scale) is plotted on a second axis, allowing users to compare the two metrics dynamically.

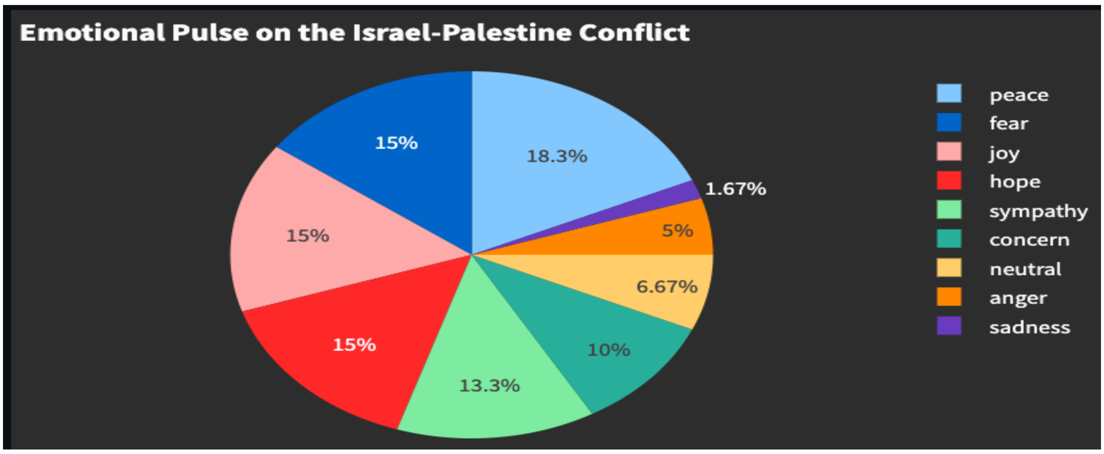


- **Trending Topics**: A table that updates in real-time, showing the latest topics being discussed on social media along with their sentiment. This helps identify the most relevant topics at any given moment.

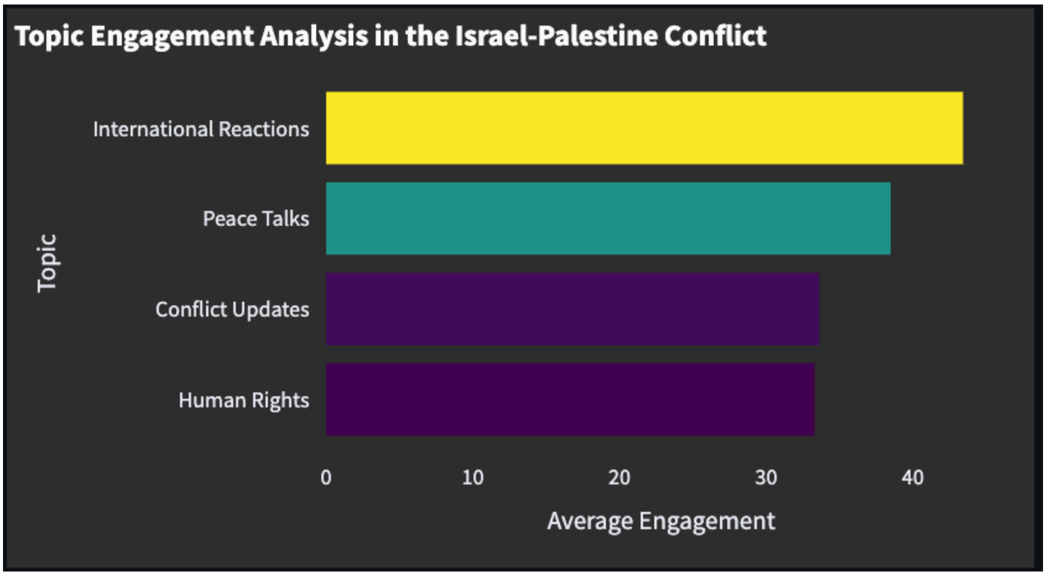

Live Trending Topics in Israel-Palestine Conflict

Time	Trending Topic	Sentiment
02:13:30	UN Security Council debate #InternationalReactions	Positive
02:13:38	International support for one side #InternationalReactions	Positive
02:13:58	Violence in Gaza #ConflictUpdates	Neutral

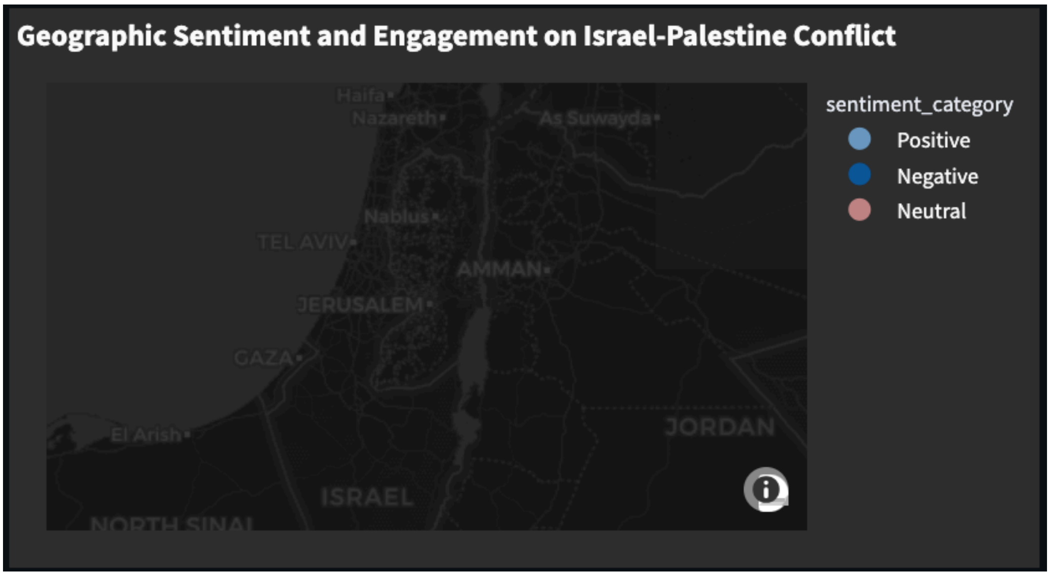
- **Emotional Pulse**: A pie chart displays the distribution of emotions in the tweets. This chart provides insight into how people are feeling about the Israel-Palestine conflict, ranging from positive emotions like hope to negative ones like anger.



● **Topic Engagement**: A horizontal bar chart that ranks the most engaging topics based on average engagement. This chart allows users to see which aspects of the conflict are generating the most discussion.



● **Geographic Sentiment Analysis**: An interactive map shows the geographic sentiment distribution, color-coded by sentiment category. This helps understand regional variations in the public's feelings about the conflict.



9. Challenges Faced

One of the main challenges in this project was ensuring the smooth operation of the real-time updates. Handling continuous data generation and updates without overwhelming the system required careful management of the session state and efficient handling of the data queue. Another challenge was ensuring that the sentiment analysis was realistic while still being randomized, which required balancing randomness and the conflict's context.

10. Conclusion

This project successfully created a real-time, interactive dashboard that provides valuable insights into social media activity surrounding the Israel-Palestine conflict. By leveraging Python, Streamlit, and Plotly, the dashboard allows users to visualize key metrics, such as sentiment, engagement, and regional sentiment distribution, in real-time. The dynamic nature of the dashboard ensures that users always have up-to-date information, which can be critical in responding to shifts in public sentiment or trending topics.

The dashboard can be used by communications teams, researchers, and anyone interested in monitoring public opinion and engagement on the Israel-Palestine conflict. The insights gained from the dashboard can help shape communication strategies, identify emerging trends, and understand the emotional landscape of the discourse.

11. Future Improvements

While the current dashboard offers a functional prototype, there are several areas for future improvements:

- **Real-World Data Integration**: Integrating real tweet data from the Twitter API would make the dashboard more relevant and accurate.
- **Scalability**: Improving the backend to handle larger datasets efficiently and optimizing the performance of real-time updates.
- **Advanced Sentiment Analysis**: Implementing more sophisticated sentiment analysis techniques, such as deep learning models, could provide more accurate sentiment predictions.

12. References

- Streamlit Documentation: https://docs.streamlit.io/
- Plotly Documentation: https://plotly.com/python/
- Python Pandas Documentation: https://pandas.pydata.org/
- Twitter API (for future real-world data integration): https://developer.twitter.com/en/docs